Retail Customer Insights and Revenue Analytics

Tools: **Python, SQL, Power BI**
Dataset Size: **3,900 Customer Transactions**

1. Project Overview

This project analyzes retail customer shopping behavior to identify patterns in purchasing trends, customer segmentation, and revenue drivers. The goal is to help the retail company to understand spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

The dataset contains transactional records from 3,900 customer purchases across multiple product categories. Python was used for data preparation, SQL for analysis, and Power BI for visualization.

2. Business Problem

A retail company wants to understand how customer demographics, product categories, and promotional strategies influence purchasing behavior and revenue generation.

The key business question is:
How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?

3. Dataset Summary

Attribute	Description
Rows	3,900
Columns	18
Customer Information	Age, Gender, Location, Subscription Status
Purchase Information	Item Purchased, Category, Frequency of Purchase, Purchase Amount, Season, Size, Color
Behavior Data	Discount Applied, Previous Purchases, Review Rating, Shipping Type
Missing Data	37 values in Review Rating column

4. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

I. **Data Loading:** Imported the dataset using pandas.

II. **Initial Exploration:** Used df.head() to show dataset, df.info() to check structure and df.describe(include="all") for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900	3900.000000	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2	NaN	6	7
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No	NaN	PayPal	Every 3 Months
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223	NaN	677	584
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN	25.351538	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN	14.447125	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN	1.000000	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN	13.000000	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN	25.000000	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN	38.000000	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN	50.000000	NaN	NaN

III. Missing Data Handling: Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

IV. Column Standardization: Renamed columns to **snake case** (i.e. from Customer ID to customer_id) for better readability and documentation.

V. Feature Engineering:

- Created **age_group** column by binning customer ages.

```
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels = labels)
```

```
df[['age', 'age_group']].head(10)
```

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior

- Created **purchase_frequency_days** column from purchase data.

```
frequency_mapping= {
   'Fortnightly': 14,
   'Weekly': 7,
   'Monthly': 30,
   'Quarterly': 90,
   'Bi-Weekly': 14,
   'Annually': 365,
   'Every 3 Months': 90}

df['purchase_frequency_days'] =
df['frequency_of_purchases'].map(frequency_mapping)
```

```
df[['purchase_frequency_days', 'frequency_of_purchases']]
```

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
...	...	...
3895	7	Weekly
3896	14	Bi-Weekly
3897	90	Quarterly
3898	7	Weekly
3899	90	Quarterly

VI. Data Consistency Check:

Verified if **discount_applied** and **promo_code_used** were redundant; dropped promo_code_used.

VII. **Database Integration:** Connected **Python script** to **MYSQL** and **loaded** the **cleaned Data Frame** into the **database** for SQL analysis.

```
[21]: from sqlalchemy import create_engine

[22]: engine = create_engine(
      "mysql+pymysql://root:7000544107#Ka@localhost:3306/customer_behavior"
      )

[23]: df.to_sql(
          name='customer',
          con=engine,
          if_exists='replace',
          index=False
      )

[23]: 3900
```

5. Data Analysis using SQL

After connecting **Python script** to **MYSQL database,** dataset **analyzed** using SQL queries to **answer key business questions:**

a) **Revenue by Gender –** Compared total revenue generated by male vs. female customers.

gender	revenue
Male	157890
Female	75191

b) **High-Spending Discount Users –** Identified customers who used discounts but still spent above the average purchase amount.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68

c) **Top 5 Products by Rating –** Found products with the highest average review ratings.

item_purchased	Average_product_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Handbag	3.78

d) Shipping Type Comparison – Compared average purchase amounts between Standard and Express shipping.

shipping_type	avg_purchase_amount
Express	60.48
Standard	58.46

e) Subscribers vs. Non-Subscribers – Compared average spend and total revenue across subscription status.

subscription_status	total_customers	avg_spend	total_revenue
Subscribed	1053	59.49	62645
Not Subscribed	2847	59.87	170436

f) Discount-Dependent Products – Identified 5 products with the highest percentage of discounted purchases.

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

g) Customer Segmentation – Classified customers into New, Returning, and Loyal segments based on purchase history.

customer_segment	number of customers
Loyal	3116
Returning	701
New	83

h) Top 3 Products per Category – Listed the most purchased products within each category.

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

i) Repeat Buyers & Subscriptions – Checked whether customers with >5 purchases are more likely to subscribe.

subscription_status	repeat_buyers
Subscribed	958
Not Subscribed	2518

j) Revenue by Age Group – Calculated total revenue contribution of each age group.

age_group	total_revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

6. Power BI Dashboard

Then connecting MYSQL to Power BI and built an interactive dashboard to present insights visually.

7. Key Insights

- Express shipping customers spend more on average.
- Loyal customers represent the majority of the customer base.
- Several products rely heavily on discounts for sales.

8. Business Recommendations

- **Boost Subscriptions –** Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs –** Reward repeat buyers to move them into the "Loyal" segment.
- **Review Discount Policy –** Optimize discount strategies to avoid over-dependence.
- **Product Positioning –** Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing –** Focus efforts on high-revenue age groups and express-shipping users.